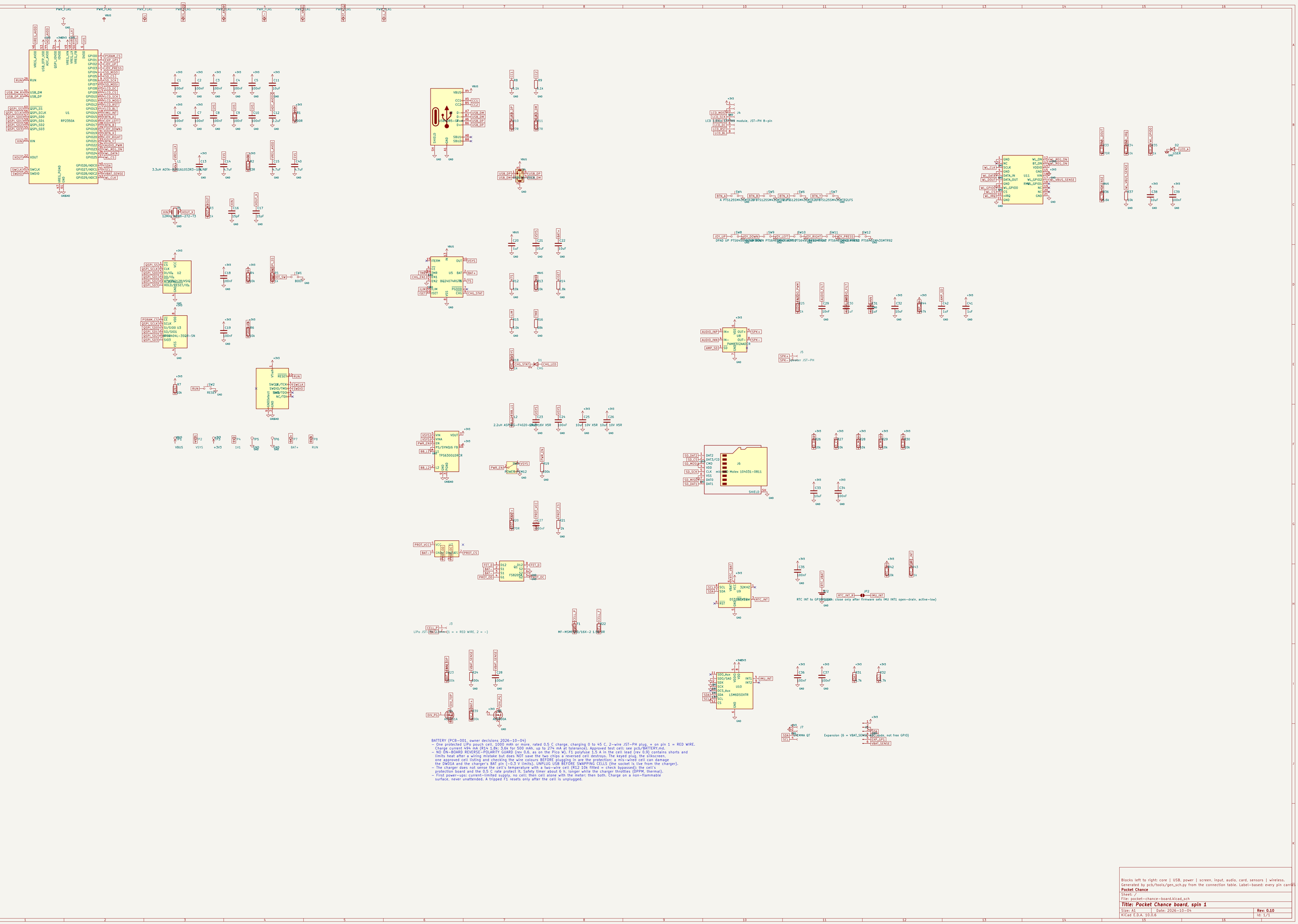

PWR_FLAG
VBUS
+3V3
GND
VREG_AVDD
VSYS
VREG_IN
VREG_LX
VREG_FB
USB_DP_MCU
USB_DM_MCU
QSPI_IOVDD
IOVDD
SWCLK
SWDIO
RUN
XIN
XOUT
USB_DM
USB_DP
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26/ADC0
GPIO27/ADC1
GPIO28/ADC2
GPIO29/ADC3
PSRAM_CS
EXP_GP1
JOY_UP
JOY_PRESS
SD_MISO
SD_CS
SD_SCK
SD_MOSI
LCD_DC
LCD_CS
LCD_SCK
LCD_MOSI
LCD_RST
LCD_BL
IMU_INT
BTN_A
JOY_LEFT
BTN_B
JOY_DOWN
BTN_X
JOY_RIGHT
BTN_Y
AUDIO_PWM
WL_REG_ON
WL_DATA
WL_CS
SDA
SCL
VBAT_SENSE
WL_CLK
RP2350A
U1
3.3uH AOTA-B201610S3R3-101-T
L1
12MHz ABM8-272-T3
X1
SWCLK
SWDIO
RUN
BOOT
BOOT_SW
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
W25Q128JVSIQ
U2
PSRAM_CS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
APS6404L-3SQR-SN
U3
VBUS
VSYS
+3V3
1V1
BAT+
RUN
SWCLK
SWDIO
RESET
SW2
TP1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
USB_DP
USB_DM
USB_CC1
USB_CC2
USB-C-SMD_HRO-TYPE-C-31-M-12
J1
SHIELD
R8
R9
5.1k
27R
R10
R11
USB_DP_MCU
USB_DM_MCU
R1
0.00R
R2
3R
R3
1k
R4
10k
R5
1k
R6
10k
R7
10k
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
C12
C13
C14
C15
C16
C17
C18
C19
C20
C21
C22
100nF
10uF
2.2uF
4.7uF
15pF
VREG_AVDD
VREG_LX
VREG_FB
VREG_IN
BOOT_SW
QSPI_SS
BQ24074RGTR
U5
ITERM
TMR
CHG_EN1
EN2
ILIM
ISET
IN
OUT
BAT
TS
PGOOD
CHG
CHG_STAT
R12
R13
R14
R15
R16
R17
10k
100k
1.8k
1.0k
68k
ILIM
ISET
TMR
D1
CHG_LED
CHG
R18
1k
C23
C24
C25
C26
2.2uH ASPI-F4020-2R2M-T
10uF 10V X5R
10uF 10V X5R
VSYS
BB_L1
BB_L2
TPS63001DRCR
U7
VIN
VOUT
EN
PS/SYNC
FB
PGND
SW
R19
100k
PWR_EN
SW4
POWER-PAD12
R20
0.00R
C27
100nF
PROT_VCC
PROT_CS
2k
PROT_VCC
PROT_CS
PROT_DO
PROT_DC
FET_D
FET_P
BAT-
D2
PMEG6010CEJ
D3
PMEG6010CEJ
Q1
FSB220
CELL_P
CELL_N
J3
LiPo JST-PH 2-pin (1 = + RED WIRE, 2 = −)
F1
MF-MSMF050/16X-2 1.5A HOLDER
CELL_P
CELL_N
R21
R22
100k
100k
R23
R24
R25
R26
100k
100k
100k
100k
VBAT_SENSE
C28
100nF
DIV_EN
DIV_PG
DIV_CPG
BAT+
DIV_CPG
DIV_EN
DIV_PG
BAT+
AP3012A
U8
AP3012A
U9
R27
R28
R29
R30
R31
R32
100k
100k
100k
100k
100k
100k
C29
C30
C31
C32
C33
C34
10nF
1uF
1uF
10uF
10uF
100nF
AUDIO_PWM
AUDIO_SCL
AUDIO_SDA
AUDIO_INT
AMP_SD
AUDIO_INP
AUDIO_INN
PAM8302AADCR
U6
SPK+
SPK−
J5
Speaker JST-PH
R33
R34
R35
R36
R37
R38
10k
10k
10k
10k
10k
10k
WL_VBUS_SENSE
WL_REG_ON
WL_CLK
WL_DATA
WL_DOUT
WL_GPIO0
WL_CS
WL_IRQ
WL_GPIO1
CYW43439
U11
WL_ON
BT_ON
VDDIO
VIN
WL_GPIO2
WL_GPIO1
C35
C36
C37
C38
C39
100nF
100nF
100nF
10uF
100nF
LCD_MOSI
LCD_SCK
LCD_DC
LCD_CS
LCD_RST
LCD_BL
LCD 1.69" 240×280 module, JST-PH 8-pin
J4
SD_DAT2
SD_CS
SD_MOSI
SD_SCK
SD_MISO
SD_DAT1
MicroSD Molex 104031-0811
J6
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
SHIELD
C40
C41
C42
C43
C44
C45
C46
C47
C48
C49
C50
C51
C52
C53
C54
100nF
1uF
1uF
100nF
100nF
100nF
100nF
100nF
100nF
100nF
100nF
100nF
100nF
100nF
BTN_A
BTN_B
BTN_X
BTN_Y
BTN_HOME
PTS125SM43SMTR92 LFS
SW4
SW5
SW6
SW7
JOY_UP
JOY_DOWN
JOY_LEFT
JOY_RIGHT
JOY_PRESS
PTS645SM43SMTR92
SW8
SW9
SW10
SW11
SW12
SDO_Aux
SDO/SA0
SDX
SCX
OCS_Aux
SDA
SCL
CS
LSM6DSOXTR
U10
INT1
INT2
IMU_INT
R39
R40
2.7k
2.7k
C55
C56
100nF
100nF
SCL
SDA
STEMMA QT
J7
Expansion (6 = VBAT_SENSE, 7 = decode, not free GPIO)
EXP_GP1
VBAT_SENSE
RTC_INT
RTC_INT_R
IMU_INT
JP2
RTC INT to GP15 (default); close only after firmware sets IMU INT1 open-drain, active-low)
BATTERY (PCB-001, owner decisions 2026-10-04)
– One protected LiPo pouch cell, 1000 mAh or more, rated 0.5 C charge, charging 0 to 45 C, 2-wire JST-PH plug, + on pin 1 = RED WIRE.
Charge current 494 mA (R14 1.8k; 3.6k for 500 mAh, up to 274 mA at tolerance). Approved test cell: see pcb/BATTERY.md.
– NO ON-BOARD REVERSE-POLARITY GUARD (rev 0.6, as on the Pico W). F1 polyfuse 1.5 A in the cell lead (rev 0.9) contains shorts and
limits heat after a wiring mistake but does NOT save the two chips a reversed cell destroys. The keyed plug, the silkscreen,
one approved cell listing and checking the wire colours BEFORE plugging in are the protection: a mis-wired cell can damage
the DW01A and the charger's BAT pin (−0.3 V limits), UNPLUG USB BEFORE SWAPPING CELLS (the socket is live from the charger).
– The charger does not sense the cell's temperature with a two-wire cell (R12 10k fitted = check bypassed): the cell's
protection board and the 0.5 C rate protect it. Safety timer about 6 h, longer while the charger throttles (@PPM, thermal).
– First power-ups: current-limited supply, no cell; then cell alone with the meter; then both. Charge on a non-flammable
surface, never unattended. A tripped F1 resets only after the cell is unplugged.
Blocks left to right: core | USB, power | screen, input, audio, card, sensors | wireless.
Generated by pcb/tools/gen_sch.py from the connection table. Label-based: every pin carries its net
Pocket Chance
Sheet: /
File: pocket-chance-board.kicad_sch
Title: Pocket Chance board, spin 1
Size: A1
Date: 2026-10-04
Rev: 0.10
KiCad E.D.A. 10.0.6
Id: 1/1